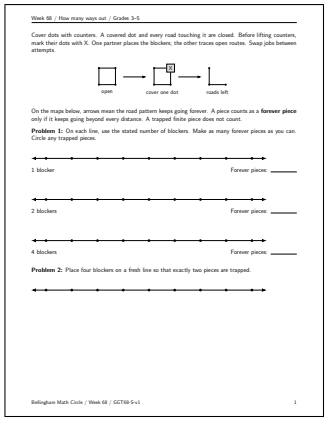
Week 68 / How many ways out / Page menu

For adults. Band slices print the pages whose range includes the band, with page and problem numbers unchanged.
K–1: no pages · Grades 2–3: whole packet · Grades 4–5: whole packet


Week 68 / How many ways out / Grades 3–5

Cover dots with counters. A covered dot and every road touching it are closed. Before lifting counters, mark their dots with X. One partner places the blockers; the other traces open routes. Swap jobs between attempts.

open cover one dot roads left

On the maps below, arrows mean the road pattern keeps going forever. A piece counts as a **forever piece** only if it keeps going beyond every distance. A trapped finite piece does not count.

Problem 1: On each line, use the stated number of blockers. Make as many forever pieces as you can. Circle any trapped pieces.

1 blocker Forever pieces: ____

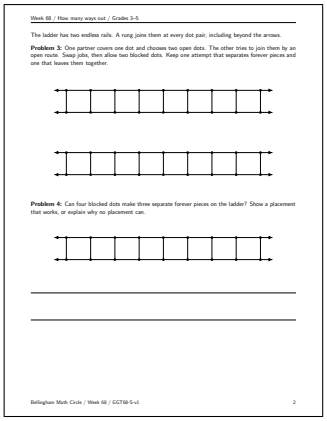
2 blockers Forever pieces: ____

4 blockers Forever pieces: ____

Problem 2: Place four blockers on a fresh line so that exactly two pieces are trapped.

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page 1 · 3–5 · P1–2


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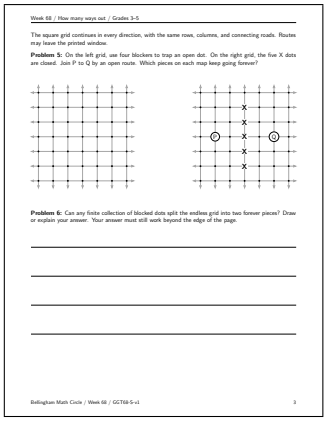
The ladder has two endless rails. A rung joins them at every dot pair, including beyond the arrows.

Problem 3: One partner covers one dot and chooses two open dots. The other tries to join them by an open route. Swap jobs, then allow two blocked dots. Keep one attempt that separates forever pieces and one that leaves them together.

Problem 4: Can four blocked dots make three separate forever pieces on the ladder? Show a placement that works, or explain why no placement can.

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page 2 · 3–5 · P3–4


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The square grid continues in every direction, with the same rows, columns, and connecting roads. Routes may leave the printed window.

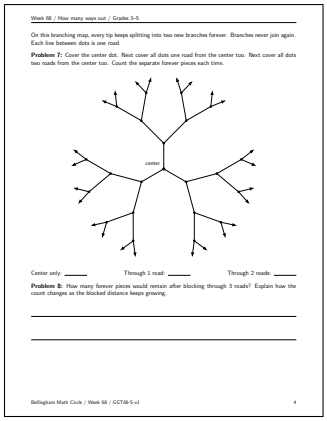
Problem 5: On the left grid, use four blockers to trap an open dot. On the right grid, the five X dots are closed. Join P to Q by an open route. Which pieces on each map keep going forever?

P Q

Problem 6: Can any finite collection of blocked dots split the endless grid into two forever pieces? Draw or explain your answer. Your answer must still work beyond the edge of the page.

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On this branching map, every tip keeps splitting into two new branches forever. Branches never join again. Each line between dots is one road.

Problem 7: Cover the center dot. Next cover all dots one road from the center too. Next cover all dots two roads from the center too. Count the separate forever pieces each time.

center

Center only: ____ Through 1 road: ____ Through 2 roads: ____

Problem 8: How many forever pieces would remain after blocking through 3 roads? Explain how the count changes as the blocked distance keeps growing.

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